Experiment Report - classification

45 configurations of 3 model(s), ranked by accuracy (higher is better) · 2026-11-11 22:39

1. Executive Summary

best configuration	RandomForestClassifier (rank #1)
accuracy	0.9492 (higher is better)
all metrics	accuracy 0.9492, precision 0.9494, recall 0.9492, f1 0.9493
hyper-parameters	{"max_depth": 8}
features	10: mean radius, mean texture, mean perimeter, mean area, mean smoothness, mean compactness, mean concavity, mean concave points, mean symmetry, mean fractal dimension
validation	k_folds
configurations / models	45 / 3
visualizations	19 generated, 4 skipped, 0 failed

- RandomForestClassifier leads LogisticRegression by 0.0117 accuracy (best configuration of each).
- LogisticRegression varies most across its configurations (std of accuracy 0.0131).
- Always predicting the most frequent class would score accuracy 0.6289; the best model scores 0.9492.
- Training time per configuration: median 0.024 s, total 13.3 s of worker time.
- The best of many configurations is optimistically biased; confirm on held-out data.

2. Experiment Configuration

created	2026-11-11 22:39:23
task	classification
validation	k_folds
ranking metric	accuracy (higher is better)
models	RandomForestClassifier, LogisticRegression, DecisionTreeClassifier
preprocessing	StandardScaler()
feature subsets tried	5
grid: max_depth	[2, 4, 8]
grid: C	[0.1, 1, 10]

3. Dataset Overview

rows	512

features	10

Feature columns: mean radius, mean texture, mean perimeter, mean area, mean smoothness, mean compactness, mean concavity, mean concave points, mean symmetry, mean fractal dimension

4. Overall Results and Model Comparison

Per model

Best, mean, spread and worst accuracy over each model's configurations, and the mean of every other metric.

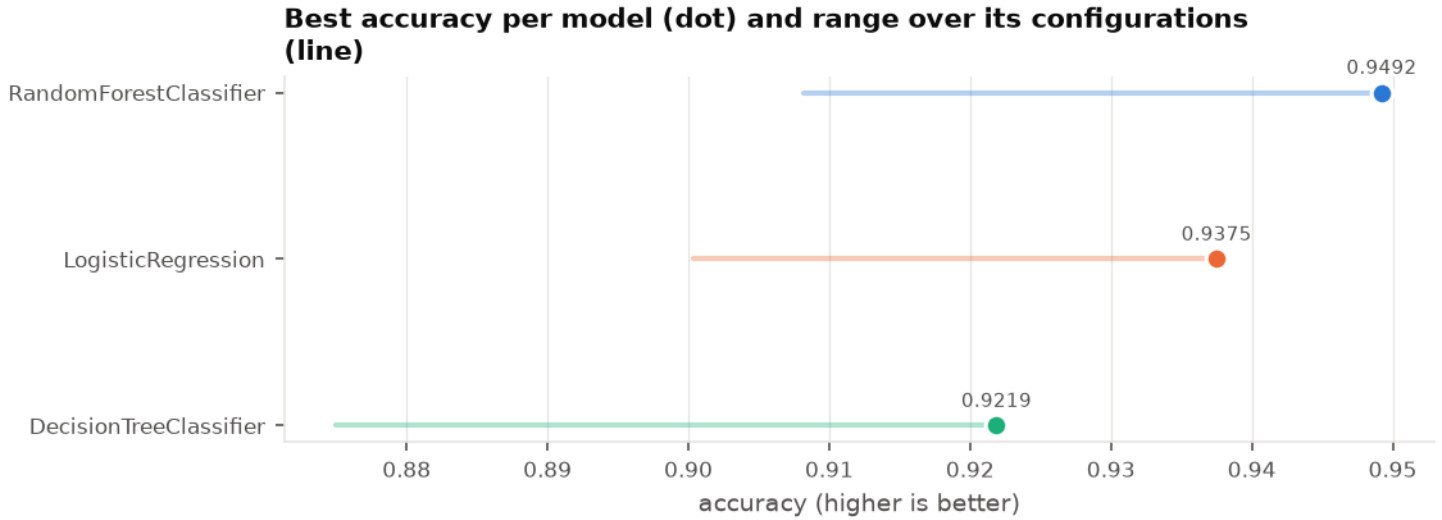
model	configurations	best_accuracy	mean_accuracy	std_accuracy	worst_accuracy	mean_precision	mean_recall	mean_f1	best_rank	best_n_features
RandomForestClassifier	15	0.9492	0.9245	0.0122	0.9082	0.9243	0.9245	0.9243	1	10
LogisticRegression	15	0.9375	0.9207	0.0131	0.9004	0.9210	0.9207	0.9201	4	8
DecisionTreeClassifier	15	0.9219	0.9066	0.0120	0.8750	0.9076	0.9066	0.9069	15	10

Top 20 configurations

rank	model	params	n_features	validation	accuracy	precision	recall	f1
1	RandomForestClassifier	{"max_depth": 8}	10	k_folds	0.9492	0.9494	0.9492	0.9493
2	RandomForestClassifier	{"max_depth": 4}	8	k_folds	0.9395	0.9393	0.9395	0.9393
3	RandomForestClassifier	{"max_depth": 8}	8	k_folds	0.9395	0.9394	0.9395	0.9394
4	LogisticRegression	{"C": 1}	8	k_folds	0.9375	0.9374	0.9375	0.9372
5	LogisticRegression	{"C": 10}	8	k_folds	0.9375	0.9374	0.9375	0.9373
6	LogisticRegression	{"C": 0.1}	10	k_folds	0.9375	0.9391	0.9375	0.9367
7	RandomForestClassifier	{"max_depth": 4}	10	k_folds	0.9355	0.9354	0.9355	0.9354
8	LogisticRegression	{"C": 1}	10	k_folds	0.9355	0.9354	0.9355	0.9353
9	LogisticRegression	{"C": 10}	10	k_folds	0.9336	0.9334	0.9336	0.9334
10	RandomForestClassifier	{"max_depth": 2}	10	k_folds	0.9316	0.9314	0.9316	0.9314
11	LogisticRegression	{"C": 0.1}	8	k_folds	0.9297	0.9306	0.9297	0.9289
12	RandomForestClassifier	{"max_depth": 4}	6	k_folds	0.9258	0.9256	0.9258	0.9256
13	RandomForestClassifier	{"max_depth": 8}	6	k_folds	0.9258	0.9256	0.9258	0.9256
14	RandomForestClassifier	{"max_depth": 2}	8	k_folds	0.9258	0.9255	0.9258	0.9255
15	DecisionTreeClassifier	{"max_depth": 4}	10	k_folds	0.9219	0.9216	0.9219	0.9217
16	RandomForestClassifier	{"max_depth": 4}	4	k_folds	0.9199	0.9197	0.9199	0.9197
17	LogisticRegression	{"C": 0.1}	4	k_folds	0.9199	0.9212	0.9199	0.9188
18	RandomForestClassifier	{"max_depth": 2}	6	k_folds	0.9199	0.9198	0.9199	0.9199
19	DecisionTreeClassifier	{"max_depth": 4}	8	k_folds	0.9199	0.9197	0.9199	0.9198
20	LogisticRegression	{"C": 0.1}	6	k_folds	0.9180	0.9194	0.9180	0.9168

Model ranking

id: model_ranking · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

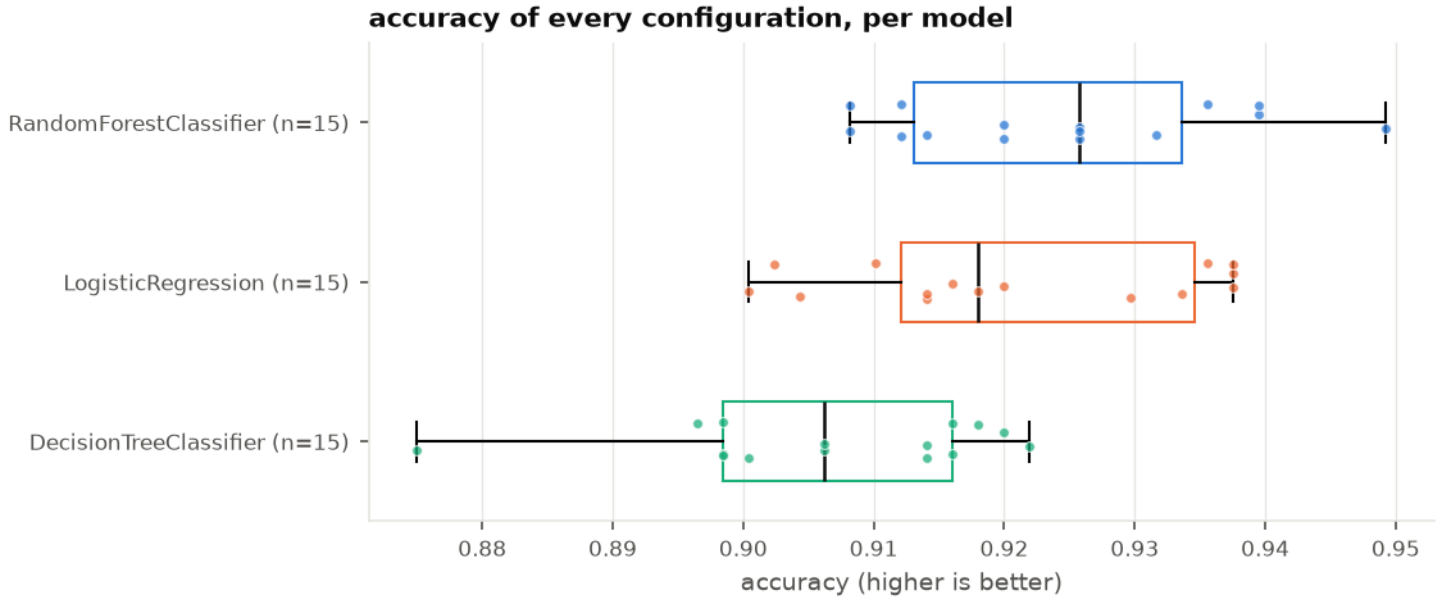


What it shows. Each model's best configuration (dot) and the range of the ranking metric over all of its configurations (line), best model on top. A long line means the model is sensitive to its hyper-parameters / feature subset. The best of many tries is optimistically biased: compare with the spread, not only the dot.

RandomForestClassifier best accuracy	0.9492
RandomForestClassifier configurations	15
LogisticRegression best accuracy	0.9375
LogisticRegression configurations	15
DecisionTreeClassifier best accuracy	0.9219
DecisionTreeClassifier configurations	15

Score distribution across configurations

id: config_distribution · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

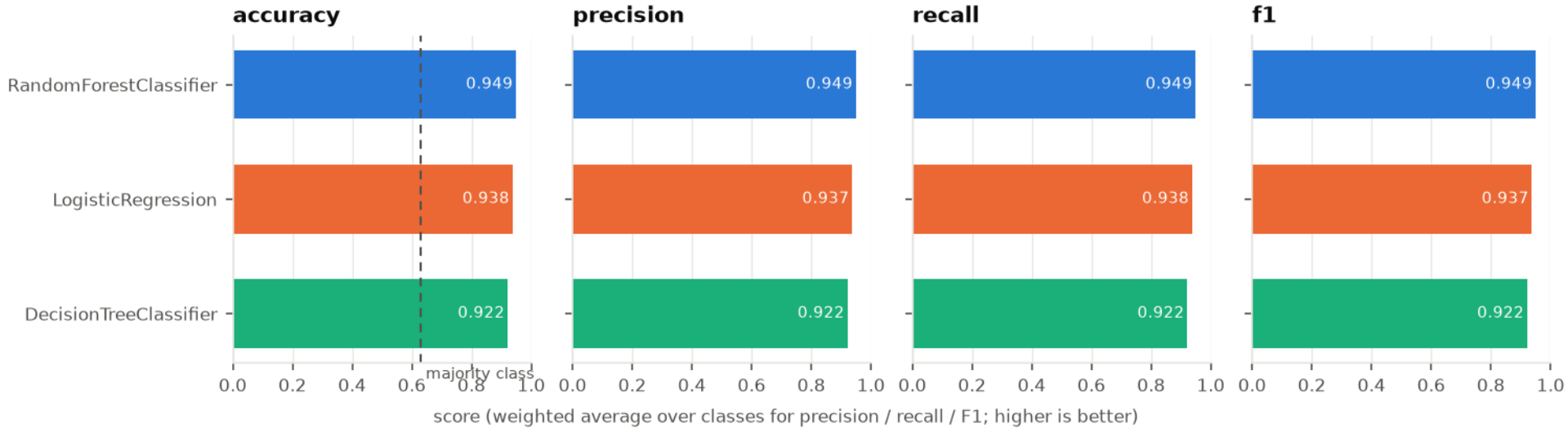


What it shows. Box plot of the ranking metric over every configuration (hyper-parameters x feature subset) of each model, with every configuration drawn as a point. Shows how robust a model family is to its settings - a narrow box near the top is a safer choice than a single lucky configuration.

RandomForestClassifier median accuracy	0.9258
LogisticRegression median accuracy	0.9180
DecisionTreeClassifier median accuracy	0.9062

Classification metrics by model

id: clf_metrics · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)



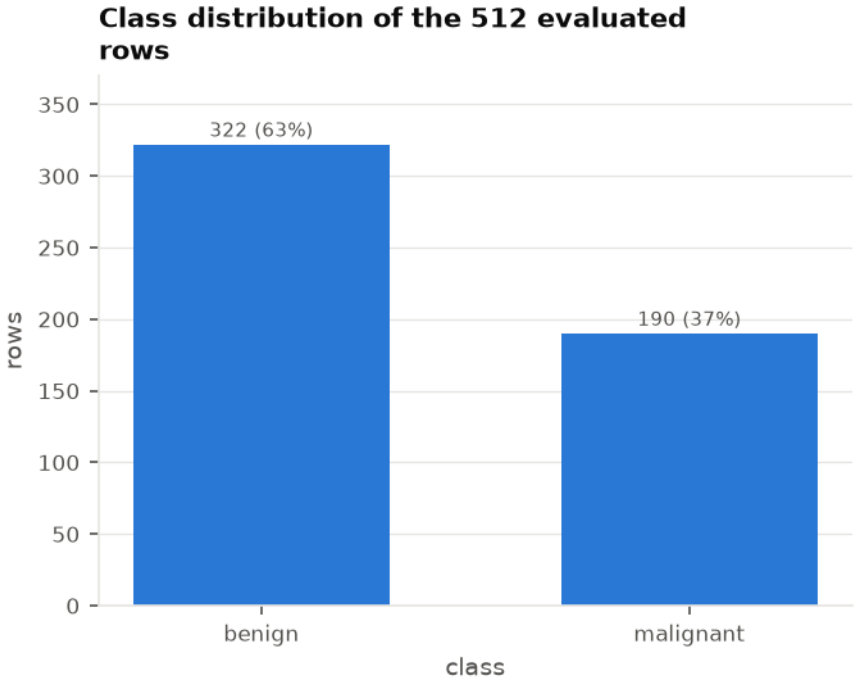
What it shows. Accuracy, precision, recall and F1 of each model's best configuration, one panel per metric (all on the same 0-1 scale). The dashed line is the accuracy of always predicting the most frequent class: a model near it has learned little.

RandomForestClassifier	accuracy 0.9492, precision 0.9494, recall 0.9492, f1 0.9493
LogisticRegression	accuracy 0.9375, precision 0.9374, recall 0.9375, f1 0.9372
DecisionTreeClassifier	accuracy 0.9219, precision 0.9216, recall 0.9219, f1 0.9217
majority-class accuracy (baseline)	0.6289

5. Classification Analysis

Class distribution

id: clf_class_distribution

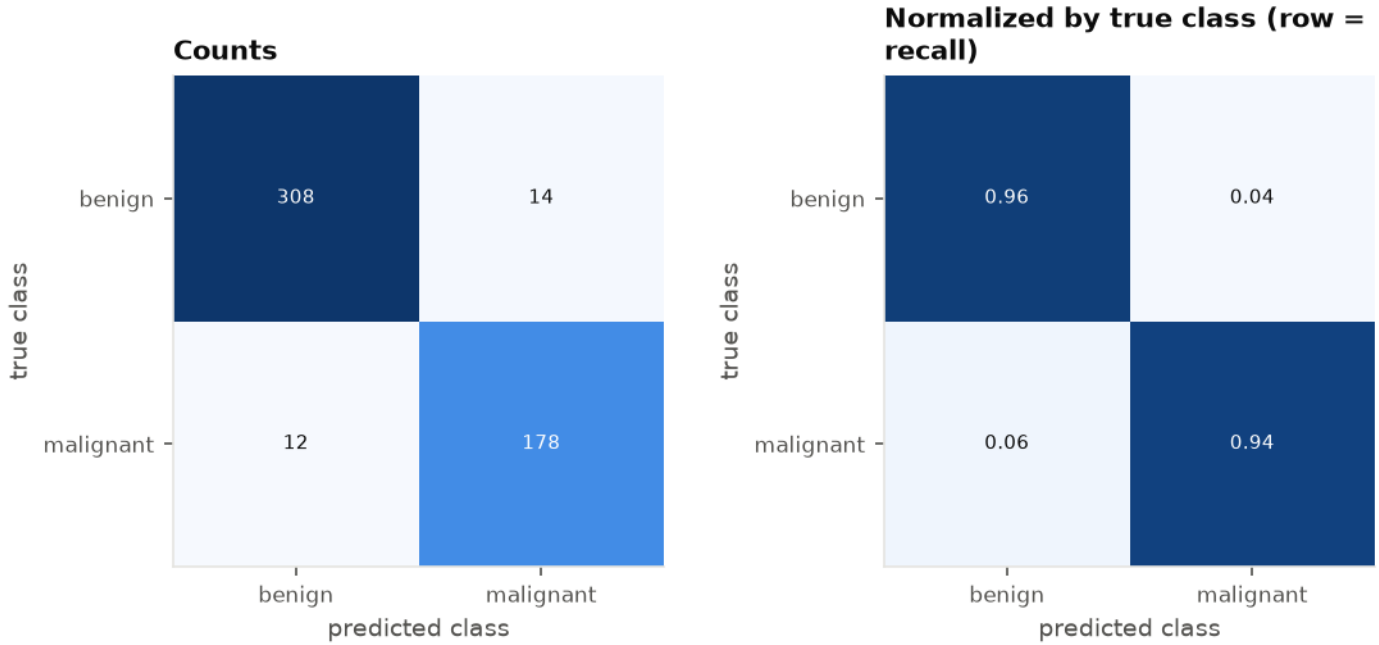


What it shows. Number of rows per class among the rows the scores were computed on (all rows for k-fold / leave-one-out, the test rows for a train / test split). Strong imbalance makes accuracy misleading - read the per-class and precision-recall plots.

classes	2
rows	512
largest / smallest class	1.69

Confusion matrix - RandomForestClassifier

id: clf_confusion:RandomForestClassifier · model(s): RandomForestClassifier (rank #1)

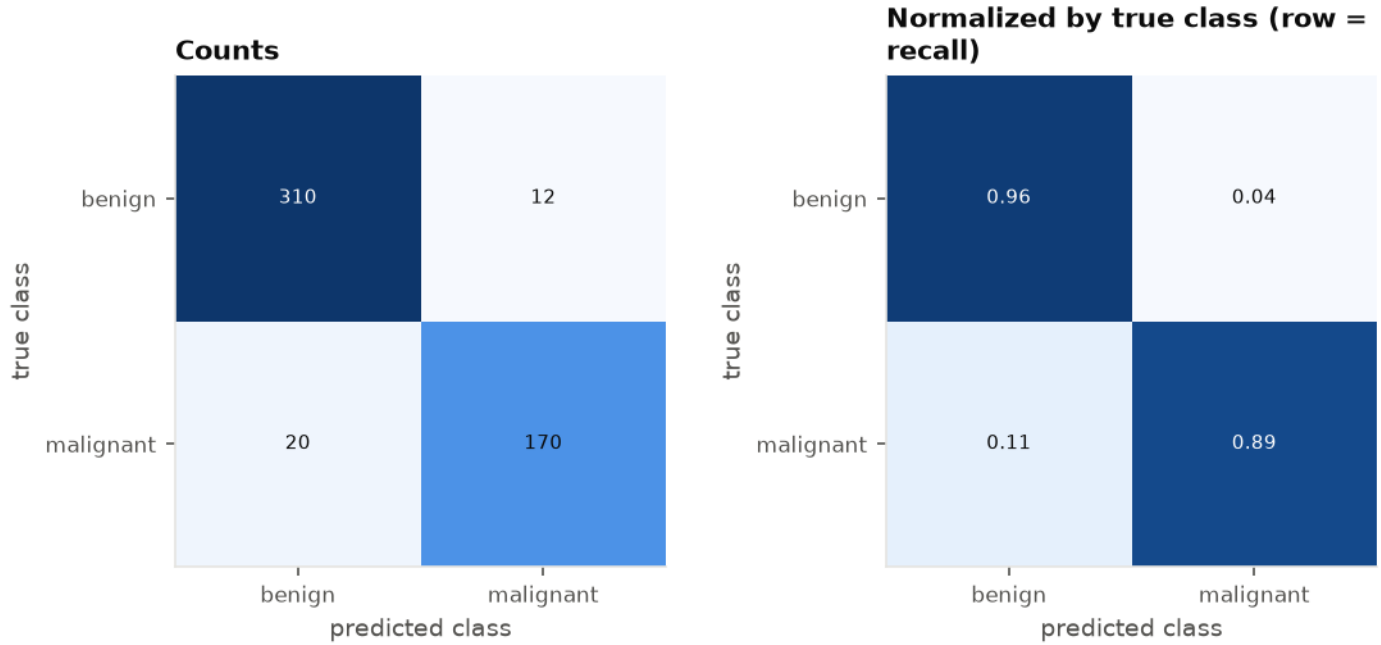


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9492
lowest recall	malignant (0.937)
rows	512

Confusion matrix - LogisticRegression

id: clf_confusion:LogisticRegression · model(s): LogisticRegression (rank #4)

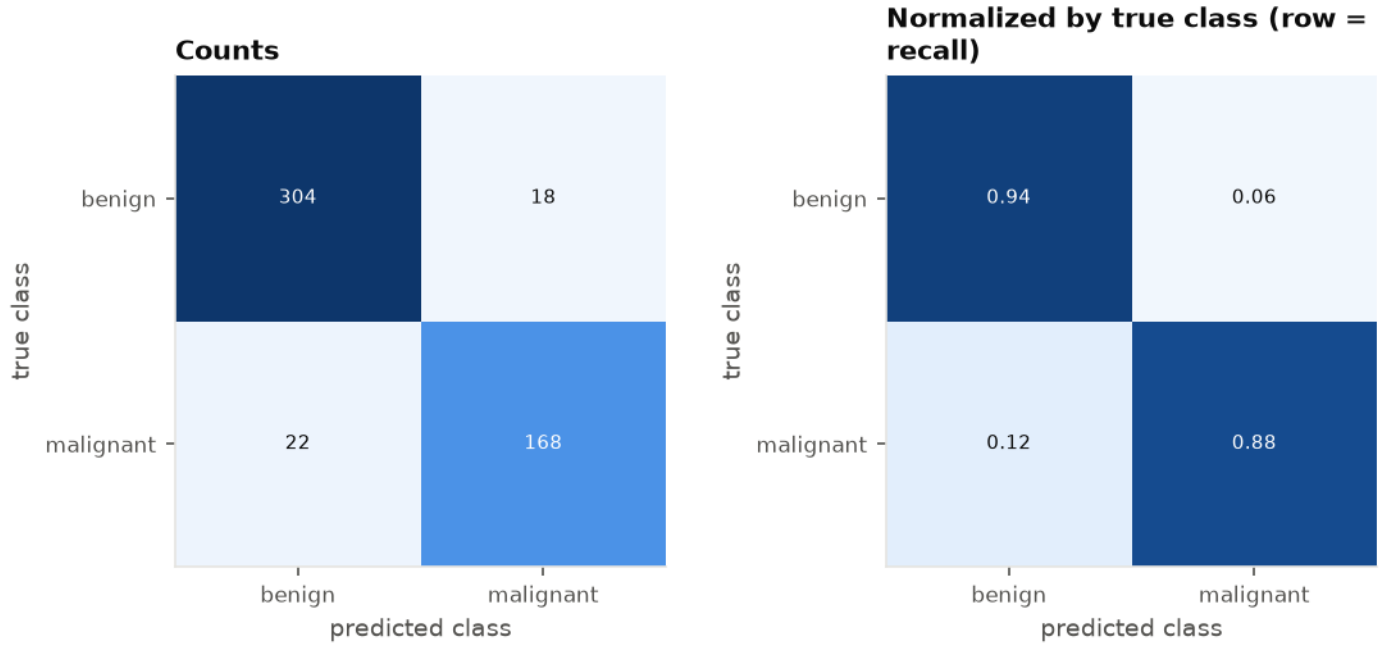


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9375
lowest recall	malignant (0.895)
rows	512

Confusion matrix - DecisionTreeClassifier

id: clf_confusion:DecisionTreeClassifier · model(s): DecisionTreeClassifier (rank #15)

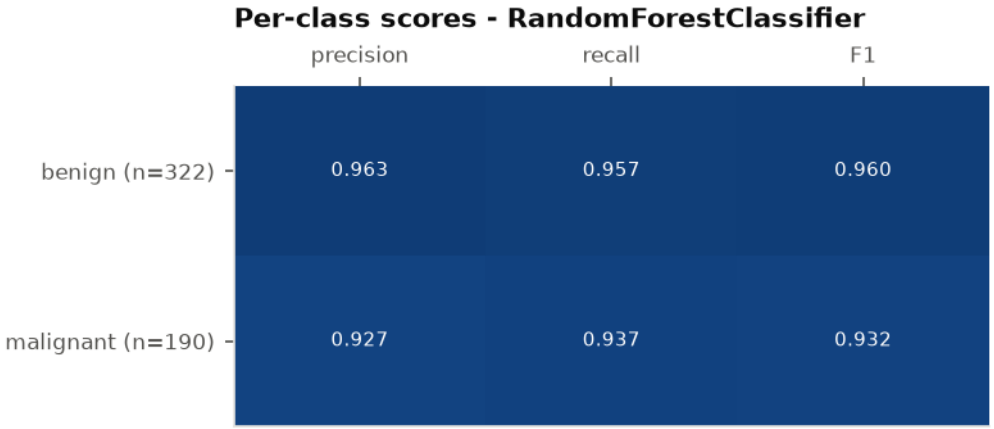


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9219
lowest recall	malignant (0.884)
rows	512

Per-class precision, recall and F1

id: clf_per_class · model(s): RandomForestClassifier (rank #1)

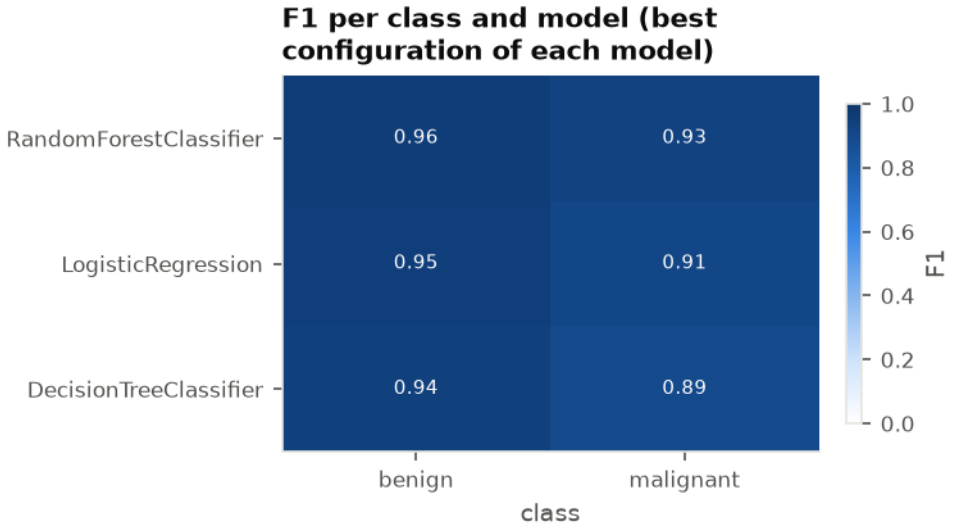


What it shows. Precision, recall and F1 of every class for the best model, with each class's support (n). The weighted averages in the leaderboard can hide a class the model almost never gets right; scores of small classes are noisy.

macro F1	0.9457
weighted F1	0.9493
worst class (F1)	malignant (0.932)

Per-class F1 across models

id: clf_per_class_models · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

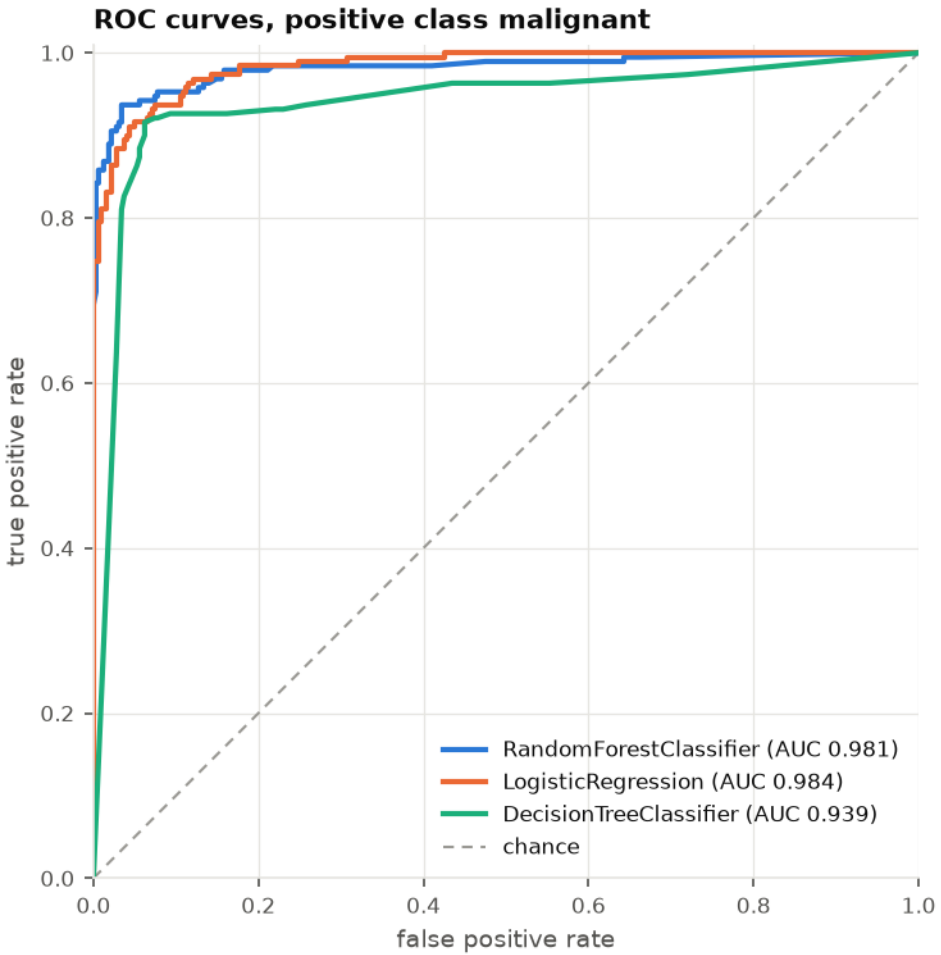


What it shows. F1 of every class for each model. Shows whether models fail on the same classes (a data problem) or on different ones (an ensemble may help).

best on benign	RandomForestClassifier
best on malignant	RandomForestClassifier

ROC curves

id: clf_roc · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

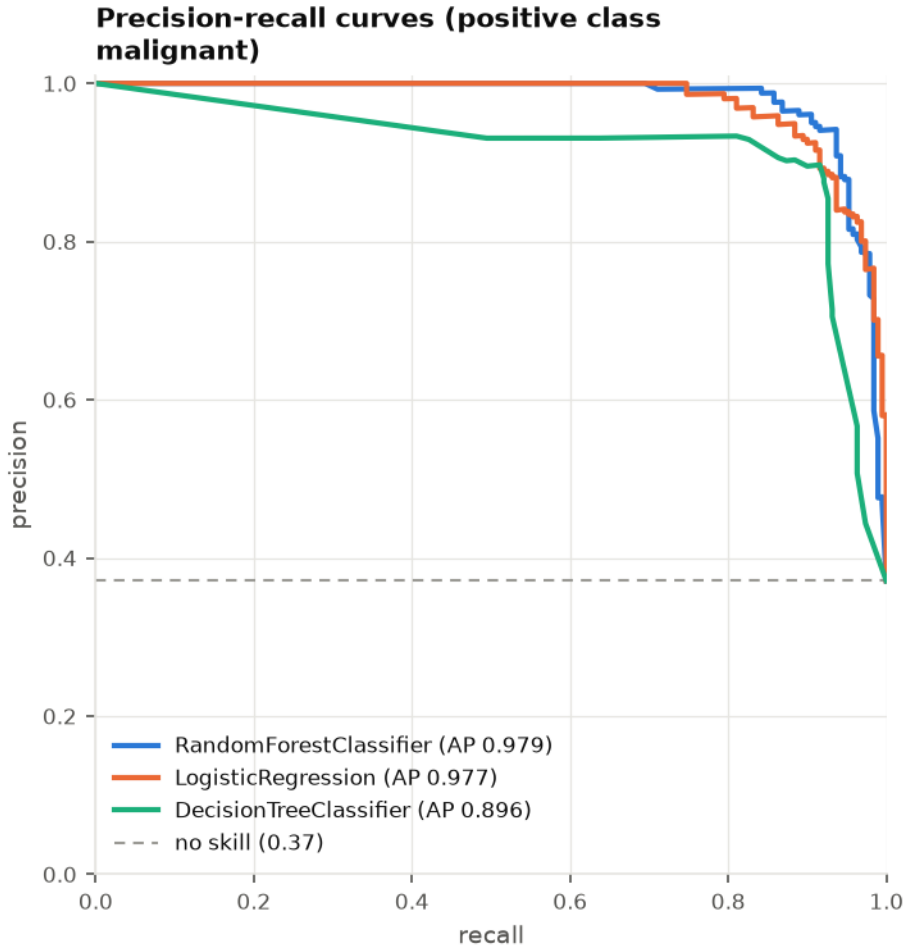


What it shows. True vs false positive rate over every decision threshold, for each model's best configuration (multiclass: micro-average over one-vs-rest classes). AUC is the probability that a random positive is ranked above a random negative; 0.5 is chance. Can look optimistic when positives are rare - see the precision-recall curves.

RandomForestClassifier ROC AUC	0.9815
LogisticRegression ROC AUC	0.9839
DecisionTreeClassifier ROC AUC	0.9389

Precision-recall curves

id: clf_pr · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

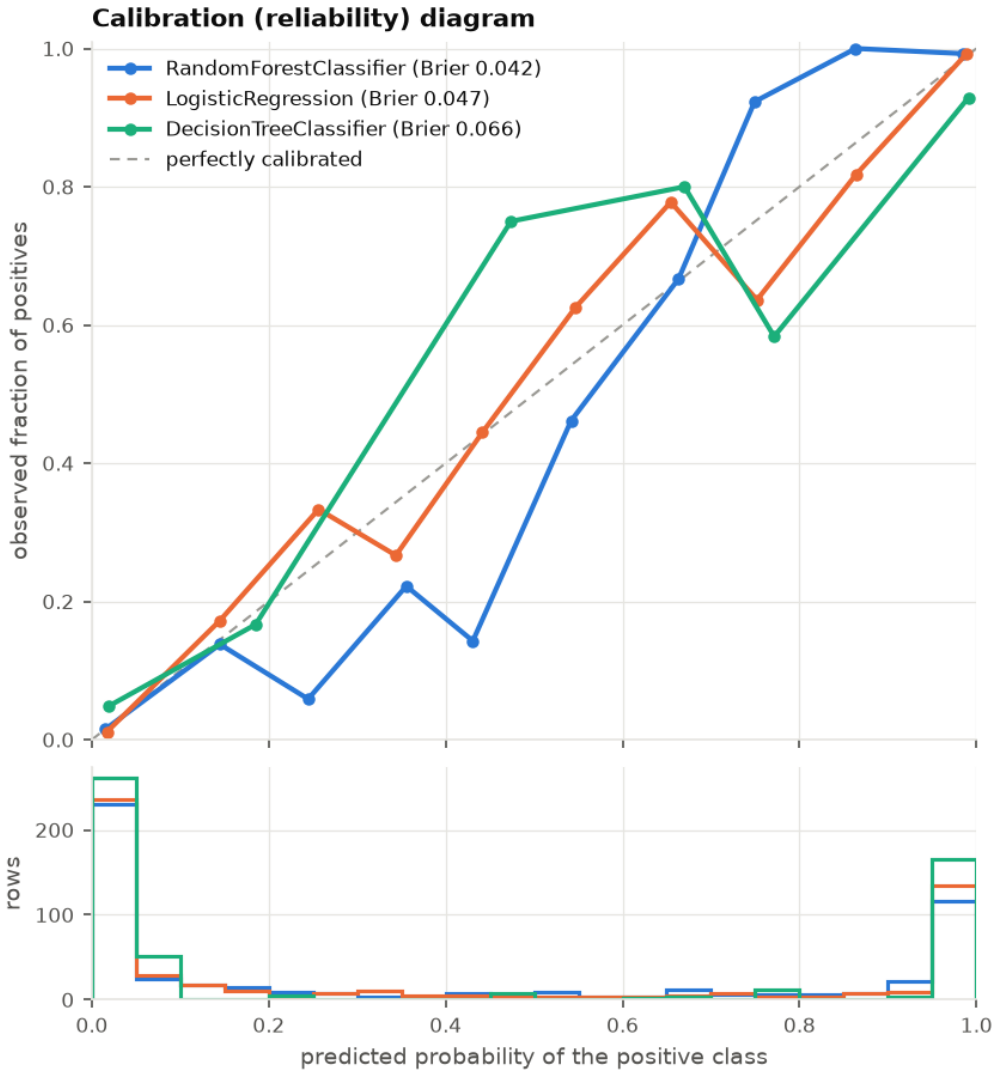


What it shows. Precision against recall over every threshold (multiclass: micro-average). Average precision (AP) summarises the curve; the dashed line is a classifier with no skill (the positive rate). More informative than ROC when the positive class is rare.

RandomForestClassifier average precision	0.9785
LogisticRegression average precision	0.9770
DecisionTreeClassifier average precision	0.8960
no-skill precision (prevalence)	0.3711

Calibration

id: clf_calibration · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)



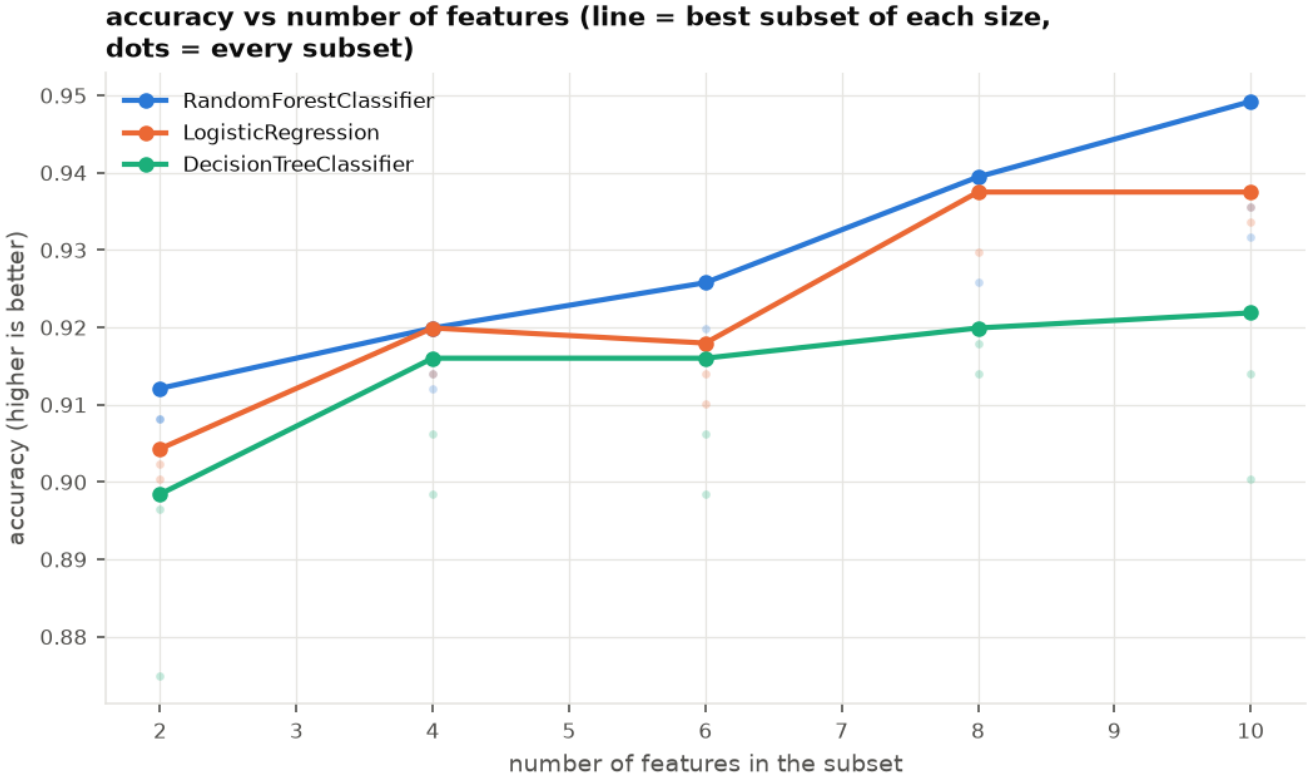
What it shows. Whether predicted probabilities can be read as probabilities: in each probability bin, the observed share of positives against the mean prediction (diagonal = perfect). Below the diagonal the model is over-confident. The histogram shows how many rows fall in each bin; the Brier score (lower is better) combines calibration and discrimination.

RandomForestClassifier Brier score	0.0418
LogisticRegression Brier score	0.0467
DecisionTreeClassifier Brier score	0.0659

6. Feature Analysis

Feature selection: score vs number of features

id: feature_count · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

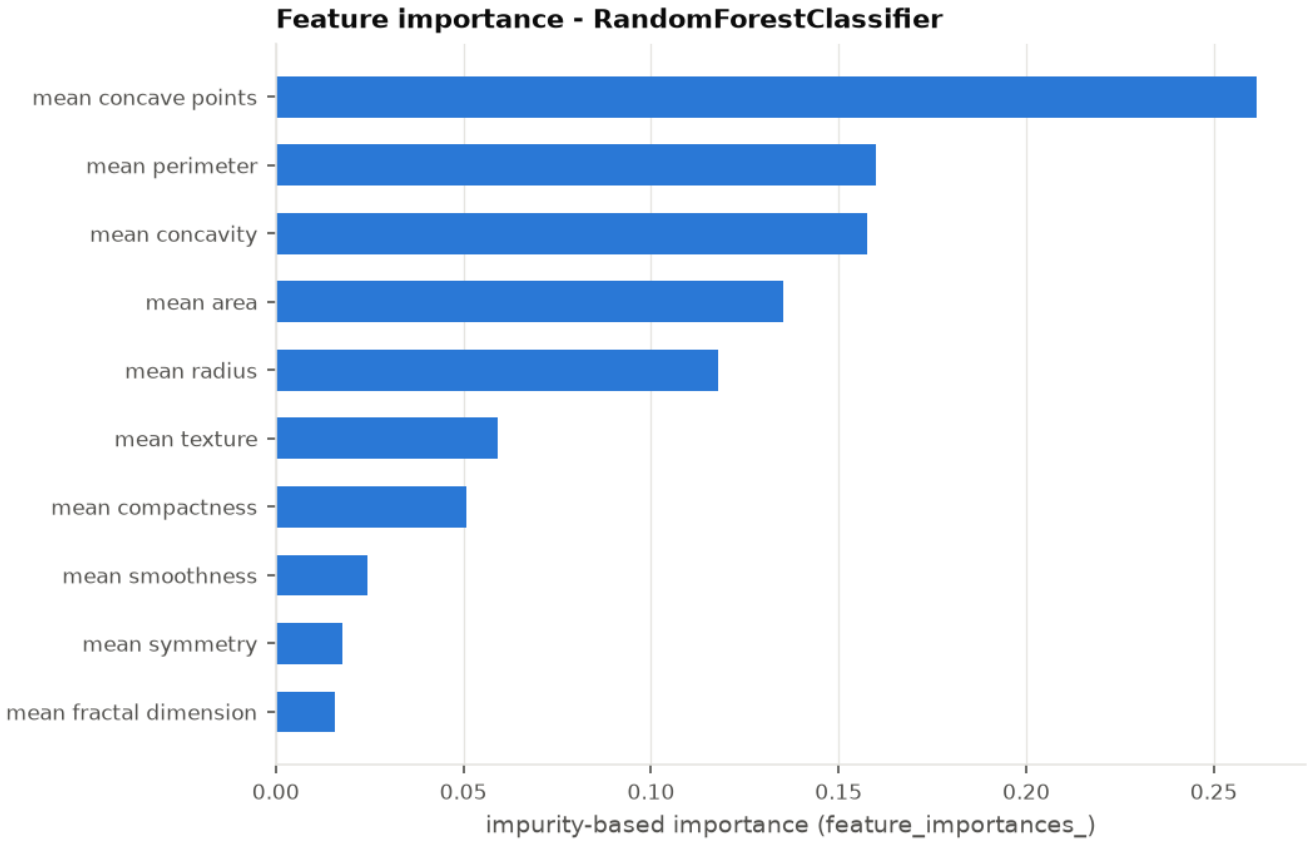


What it shows. Performance of the feature subsets tried (brute force, SelectKBest or forward / backward selection) against their size. Where the best-of-size line flattens, extra features stop paying off - prefer the smaller subset.

RandomForestClassifier best subset size	10 features (accuracy 0.9492)
LogisticRegression best subset size	8 features (accuracy 0.9375)
DecisionTreeClassifier best subset size	10 features (accuracy 0.9219)

Feature importance - RandomForestClassifier

id: feature_importance:RandomForestClassifier · model(s): RandomForestClassifier (rank #1)



What it shows. The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

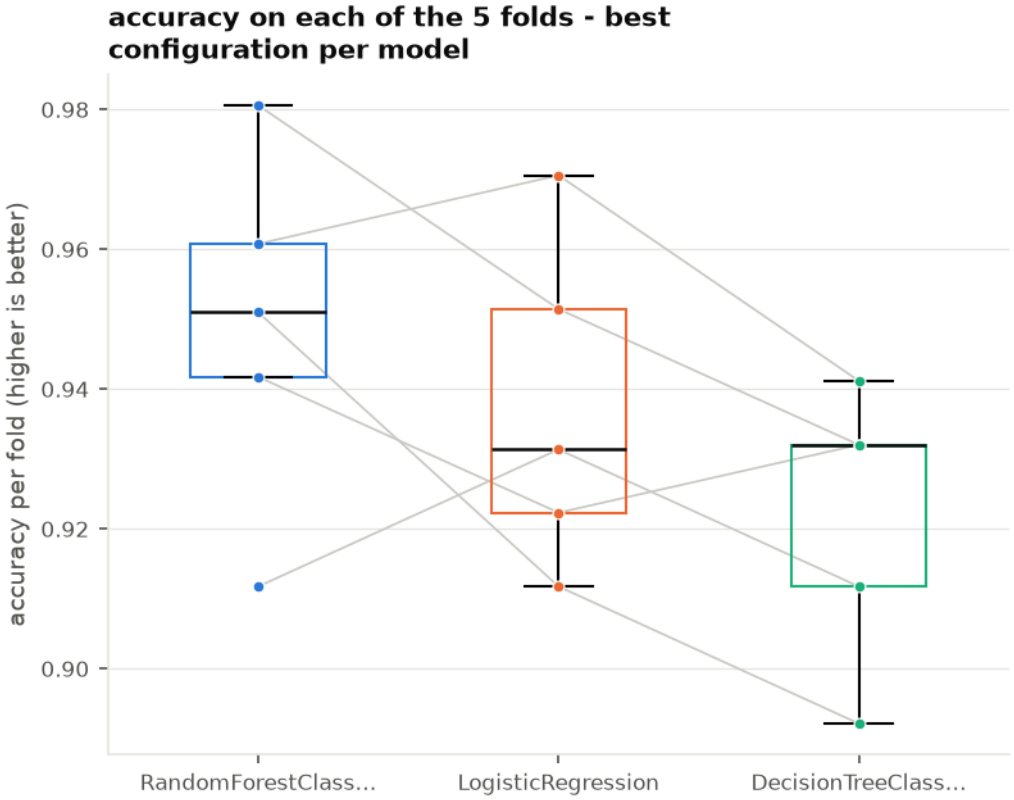
Impurity importance favours continuous / high-cardinality features and is computed on training data.

mean concave points	0.2613
mean perimeter	0.1599
mean concavity	0.1578
mean area	0.1352
mean radius	0.1180
mean texture	0.0591
mean compactness	0.0508
mean smoothness	0.0243
mean symmetry	0.0178
mean fractal dimension	0.0159

7. Cross-Validation Analysis

Metric distribution across cross-validation folds

id: cv_folds · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)



What it shows. The ranking metric computed separately on each validation fold for each model's best configuration. If the gap between two models is small compared with the fold-to-fold spread, the ranking is not reliable. Folds share training rows, so the spread understates the true variance and no significance test is implied.

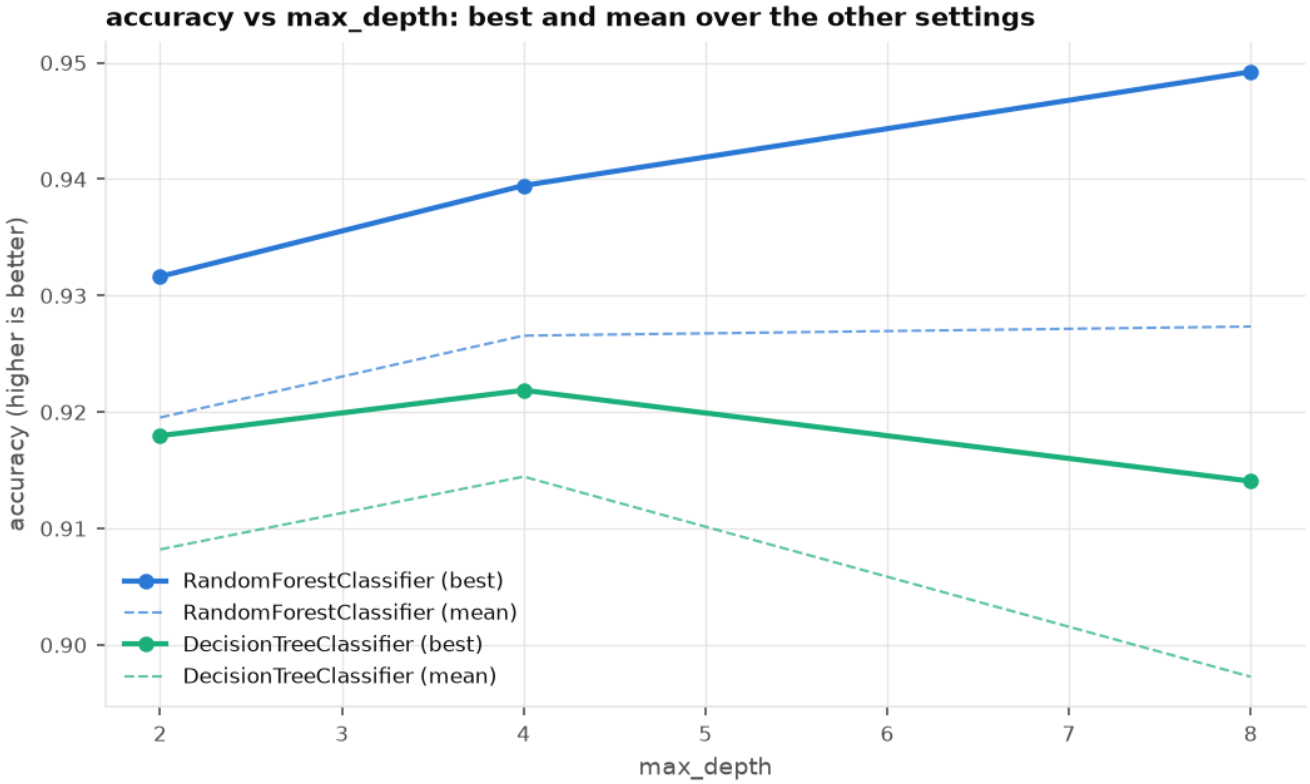
Grey lines join the same fold across models (every model is scored on identical folds).

RandomForestClassifier accuracy mean ± sd	0.9492 ± 0.0254
LogisticRegression accuracy mean ± sd	0.9375 ± 0.0236
DecisionTreeClassifier accuracy mean ± sd	0.9218 ± 0.0198

8. Hyperparameter Optimization

Hyper-parameter sensitivity - max_depth

id: hp_sensitivity:max_depth · model(s): RandomForestClassifier (rank #1), DecisionTreeClassifier (rank #15)

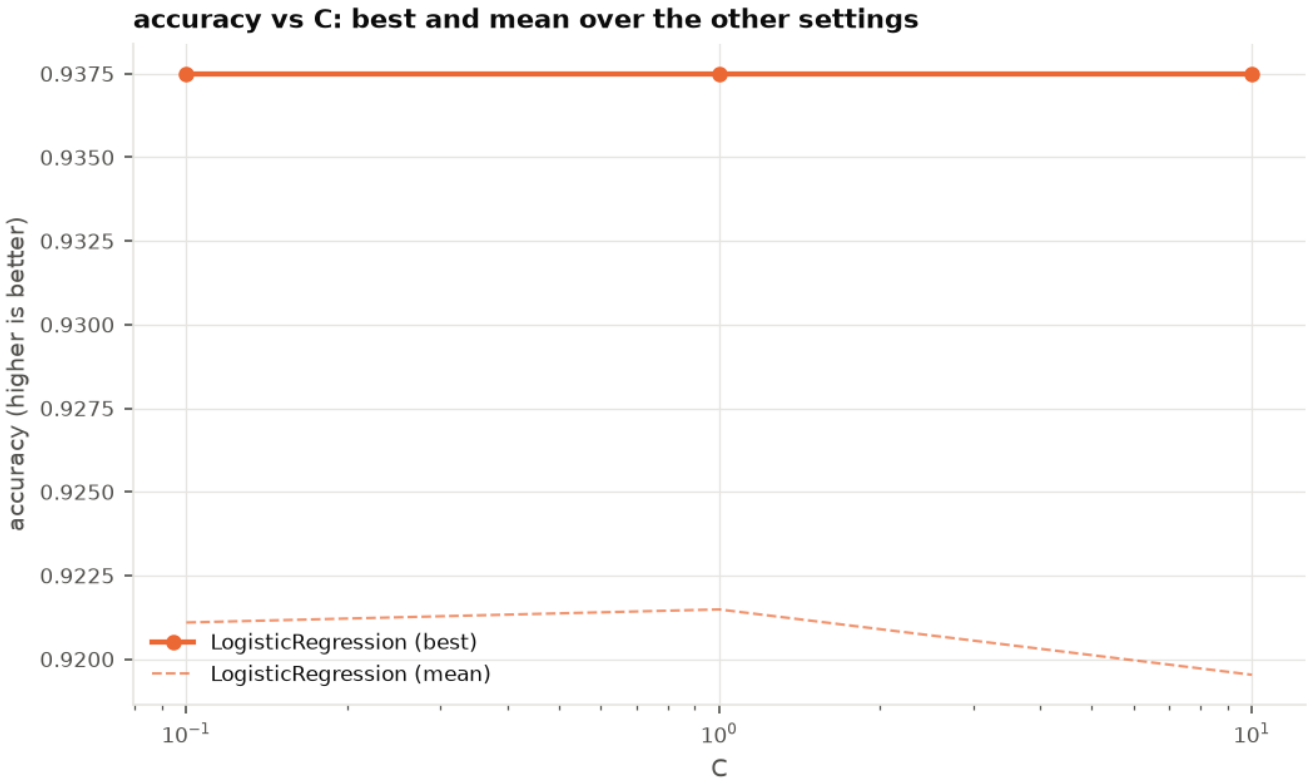


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

RandomForestClassifier best max_depth	8 (accuracy 0.9492)
DecisionTreeClassifier best max_depth	4 (accuracy 0.9219)

Hyper-parameter sensitivity - C

id: hp_sensitivity:C · model(s): LogisticRegression (rank #4)



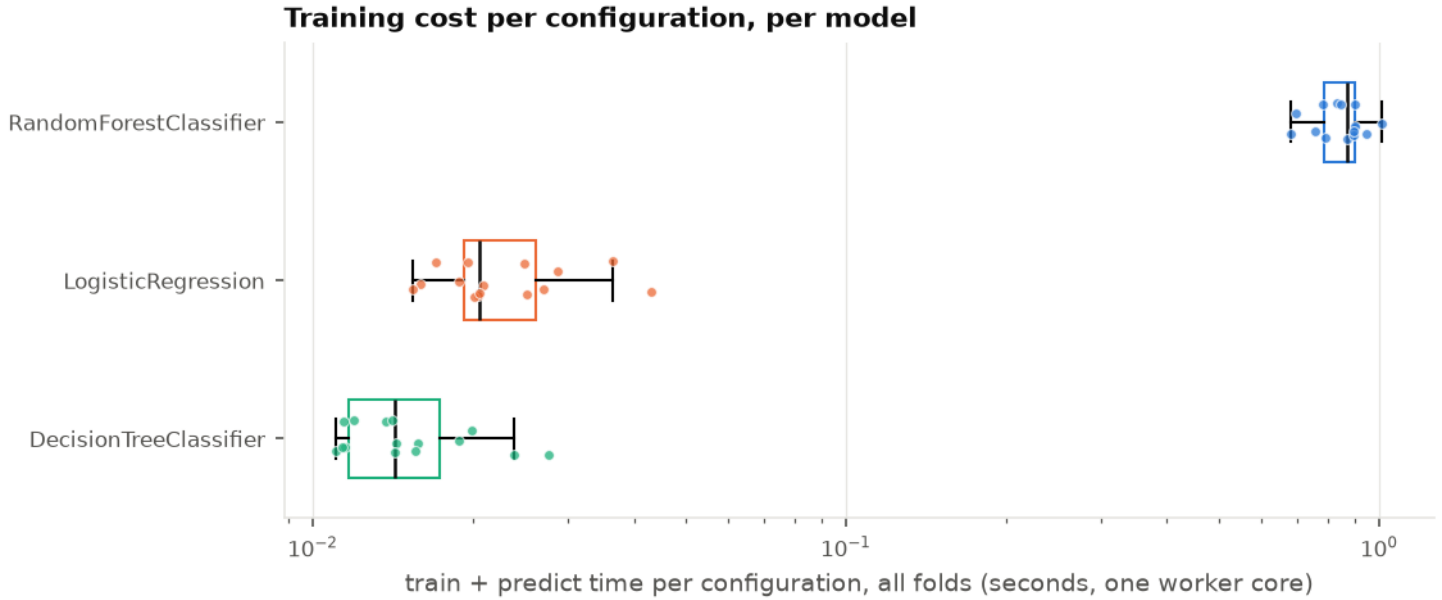
What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

LogisticRegression best C	0.1 (accuracy 0.9375)

9. Runtime and Resource Analysis

Runtime comparison

id: runtime · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)

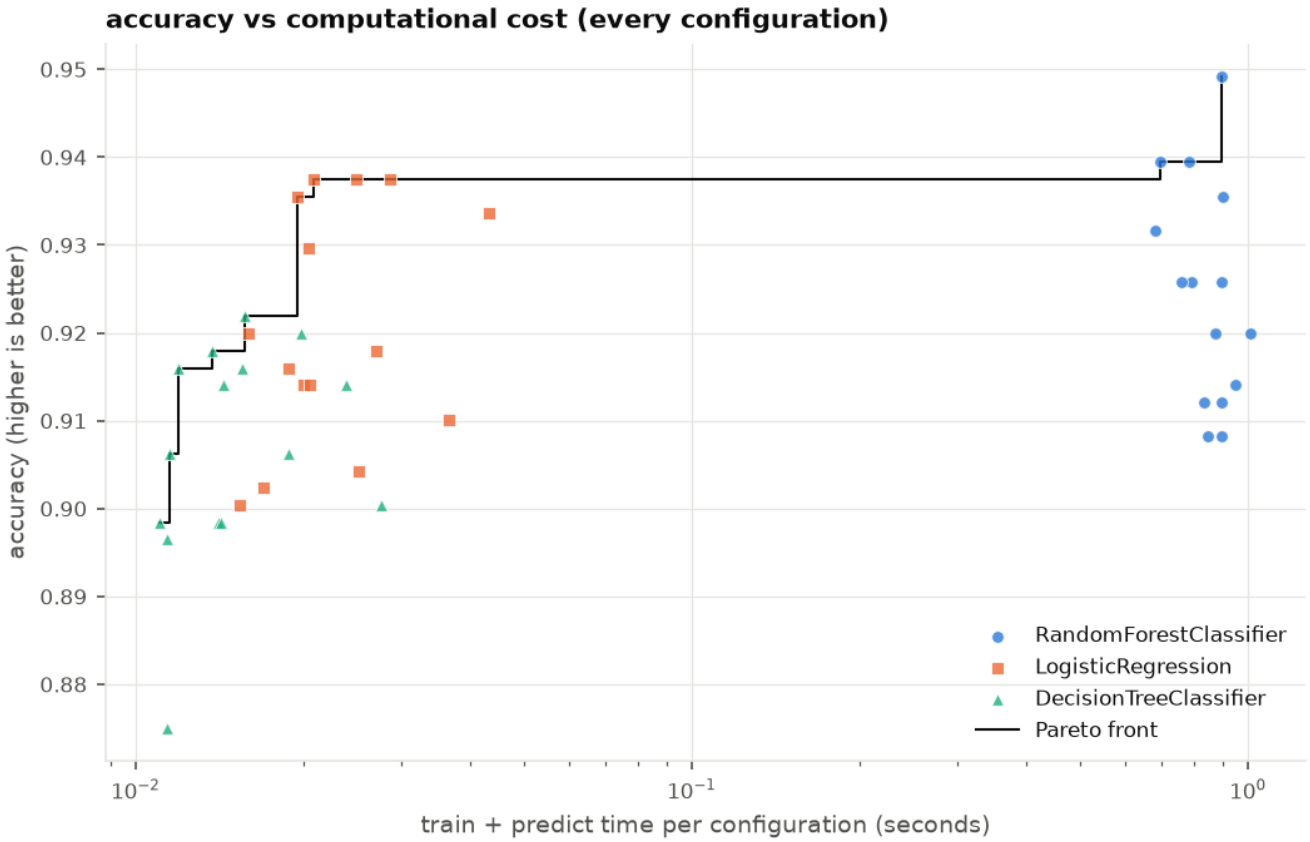


What it shows. Wall-clock time of each configuration (fit and predict over every fold), measured inside the worker process on one core. Includes preprocessing inside the pipeline; excludes data transfer. Absolute values depend on the machine and its load; compare models within one run.

RandomForestClassifier median / total	0.872 s / 12.71 s
LogisticRegression median / total	0.021 s / 0.35 s
DecisionTreeClassifier median / total	0.014 s / 0.24 s

Performance vs computational cost

id: cost_vs_score · model(s): RandomForestClassifier (rank #1), LogisticRegression (rank #4), DecisionTreeClassifier (rank #15)



What it shows. Every configuration's score against its training time. The Pareto front joins the configurations that no other one beats on both score and time: a configuration near the top of the front but much further left is almost as good for a fraction of the cost.

configurations on the Pareto front	9
fastest configuration	0.011 s (accuracy 0.8984)
best configuration	0.898 s (accuracy 0.9492)

10. Visualization Index

Every visualization attempted for this report: 19 generated, 4 skipped, 0 failed.

#	visualization	section	status	time (s)	detail
1	Model ranking	Overall Results and Model Comparison	SUCCESS	0.09	included in the report
2	Score distribution across configurations	Overall Results and Model Comparison	SUCCESS	0.10	included in the report
3	Metric distribution across cross-validation folds	Cross-Validation Analysis	SUCCESS	0.14	included in the report
4	Hyper-parameter sensitivity - max_depth	Hyperparameter Optimization	SUCCESS	0.13	included in the report
5	Hyper-parameter sensitivity - C	Hyperparameter Optimization	SUCCESS	0.15	included in the report
6	Two-parameter interaction	Hyperparameter Optimization	SKIPPED	0.00	no model has two hyper-parameters with more than one value
7	Feature selection: score vs number of features	Feature Analysis	SUCCESS	0.15	included in the report
8	Runtime comparison	Runtime and Resource Analysis	SUCCESS	0.12	included in the report
9	Performance vs computational cost	Runtime and Resource Analysis	SUCCESS	0.18	included in the report
10	Memory consumption comparison	Runtime and Resource Analysis	SKIPPED	0.00	memory consumption is not measured per configuration by the pipeline; no values are estimated
11	Classification metrics by model	Overall Results and Model Comparison	SUCCESS	0.22	included in the report
12	Class distribution	Classification Analysis	SUCCESS	0.08	included in the report
13	Confusion matrix - RandomForestClassifier	Classification Analysis	SUCCESS	0.11	included in the report
14	Confusion matrix - LogisticRegression	Classification Analysis	SUCCESS	0.10	included in the report
15	Confusion matrix - DecisionTreeClassifier	Classification Analysis	SUCCESS	0.11	included in the report
16	Per-class precision, recall and F1	Classification Analysis	SUCCESS	0.12	included in the report
17	Per-class F1 across models	Classification Analysis	SUCCESS	0.10	included in the report
18	ROC curves	Classification Analysis	SUCCESS	0.13	included in the report
19	ROC per class	Classification Analysis	SKIPPED	0.00	binary target: the single curve is in the model comparison plot
20	Precision-recall curves	Classification Analysis	SUCCESS	0.12	included in the report
21	Precision-recall per class	Classification Analysis	SKIPPED	0.00	binary target: the single curve is in the model comparison plot
22	Calibration	Classification Analysis	SUCCESS	0.20	included in the report
23	Feature importance - RandomForestClassifier	Feature Analysis	SUCCESS	0.13	included in the report

11. Errors, Warnings and Missing Visualizations

Skipped visualizations (4)

Not applicable to the data or results available - nothing was estimated to fill the gap.

Two-parameter interaction: no model has two hyper-parameters with more than one value

Memory consumption comparison: memory consumption is not measured per configuration by the pipeline; no values are estimated

ROC per class: binary target: the single curve is in the model comparison plot

Precision-recall per class: binary target: the single curve is in the model comparison plot